



**NephroSasa Rwanda: Longitudinal Artificial Intelligence-Powered
Pre-Dialysis Kidney Health Monitoring Platform for
Hypertensive and Type 2 Diabetic Patients**

BSc. in Software Engineering

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Chronic Kidney Disease affects an estimated 850 million people worldwide and is among the fastest-growing causes of premature death globally, with low- and middle-income countries bearing a disproportionate share of this burden (Lancet Global Health, 2024). Globally, hypertension and Type 2 diabetes are the two leading causes of Chronic Kidney Disease, collectively responsible for over 70% of all cases (Lancet Global Health, 2024). In Sub-Saharan Africa, the situation is particularly severe: the region has the lowest nephrology workforce density in the world and the weakest early-detection infrastructure, meaning the majority of patients are only identified at advanced, irreversible stages of the disease (ISN Global Kidney Health Atlas, 2024). In Rwanda specifically, both hypertension and Type 2 diabetes are rising at an alarming rate. Rwanda's 2024 Vital Statistics Report confirms that non-communicable diseases now account for 47.7% of all hospital deaths in the country (Rwanda National Institute of Statistics, 2024). Between 2020 and 2023, hypertension cases in Rwanda grew from 58,000 to 112,000, nearly doubling in three years.

Rwanda has a 16.8% adult hypertension prevalence nationwide and a 2.9% Type 2 diabetes prevalence, rising to 9.8% in Kigali (World Health Organization, current). Rwanda operates integrated Non-Communicable Disease clinics across all 30 districts. Stable patients with hypertension or Type 2 diabetes attend these clinics once every three months. Their blood pressure and blood sugar are recorded at each visit. However, no existing platform connects those readings across visits to detect whether kidney function is silently declining between appointments (Assessment of Blood Pressure Control Among Hypertensive Patients, 2025; Nkurunziza et al., 2025). The Assessment of Blood Pressure Control Among Hypertensive Patients Attending Rwandan District Hospital Non-Communicable Disease Clinics (2025) explicitly confirms that blood pressure readings are captured at each visit but are never analyzed longitudinally for kidney risk trajectory, and Nkurunziza et al. (2025) identified this monitoring gap as the central structural failure in Rwanda's kidney care system.

A study conducted between May 2023 and March 2024 across Rwanda's three main referral hospitals, King Faisal Hospital, the Centre Hospitalier Universitaire de Kigali, and Rwanda Military Hospital, found that at least 240 new advanced Chronic Kidney

Disease cases appear every year, with the study explicitly calling for preventive measures and early intervention as the central gaps in Rwanda's kidney care system (Nkurunziza et al., 2025). When patients are finally detected, they are already at Stage 4 or Stage 5 Chronic Kidney Disease, at which point dialysis is the only remaining treatment option. Dialysis costs RWF 75,000 per session, three times a week, approximately RWF 900,000 per month for an uninsured patient (Cost Effectiveness and Resource Allocation, 2024). No insurance scheme in Rwanda currently covers preventative kidney monitoring for patients in the Stage 1 to Stage 3a window, the exact window where intervention is clinically possible and financially affordable (BMC Nephrology, 2022).

NephroSasa Rwanda, where Sasa means "Now" in Swahili, is a longitudinal, pre-symptomatic, pre-dialysis kidney health monitoring platform designed specifically for Rwandan adults already diagnosed with hypertension or Type 2 diabetes. The platform's tagline is: Monitor Your Kidneys. Before It's Too Late.

1.2 Problem Statement

Rwanda's integrated Non-Communicable Disease clinic system successfully brings hypertensive and Type 2 diabetic patients into regular contact with healthcare providers across all 30 districts. However, the system captures readings in isolation. A nurse records a patient's blood pressure at a visit in January, and again in March, and again in June, but no platform connects those readings across time to evaluate whether the patient's kidney function is quietly declining (Assessment of Blood Pressure Control Among Hypertensive Patients, 2025). The Assessment of Blood Pressure Control Among Hypertensive Patients Attending Rwandan District Hospital Non-Communicable Disease Clinics (2025) confirms that these quarterly readings are documented in isolation, with no system performing cross-visit trajectory analysis. Clinically, this matters because Chronic Kidney Disease progresses silently over months and years: a single blood pressure reading of 145/90 mmHg tells a clinician very little on its own, but the same reading appearing consistently across six consecutive quarterly visits, with no downward trend, is a warning signal for progressive kidney stress (Lancet Global Health, 2024). Each visit sees a snapshot. No system sees the trajectory. Nkurunziza et al. (2025) directly attribute Rwanda's late-stage detection pattern to this structural gap,

noting that patients arrive at referral hospitals already at Stage 4 or Stage 5, precisely because no monitoring system detected the decline during the years it was occurring.

The consequences of this gap are severe and quantifiable. At least 240 new advanced Chronic Kidney Disease cases appear annually across Rwanda's three main referral hospitals alone (Nkurunziza et al., 2025). The mean age of these patients is 52 years, meaning the silent progression from hypertension or diabetes into kidney failure typically occurs between ages 30 and 50, precisely the most economically productive period of a Rwandan adult's life (Nkurunziza et al., 2025). Each of these patients requires dialysis at RWF 75,000 per session, three times per week, totaling RWF 900,000 per month for uninsured patients, a cost that exceeds the average monthly income of most Rwandan households (Cost Effectiveness and Resource Allocation, 2024). For patients covered by RAMA, a co-payment of approximately RWF 135,000 per month remains after insurance, and Mutuelle de Sante covers dialysis for only six weeks in acute cases (BMC Nephrology, 2022). For uninsured patients, the full RWF 900,000 monthly burden is financially unsurvivable. These are not abstract consequences. They are measurable financial catastrophes that could be avoided entirely if kidney decline were detected during the silent progression window, before Stage 4 or Stage 5 is reached. That is precisely the window NephroSasa is designed to monitor.

No existing software platform in Rwanda or at African Leadership University addresses this specific gap: longitudinal kidney health monitoring for hypertensive and Type 2 diabetic patients before symptoms appear and before a formal Chronic Kidney Disease diagnosis is made (Niyonzima et al., 2024). This work introduces NephroSasa Rwanda, a three-layer platform that combines unified longitudinal health records, an artificial intelligence Chronic Kidney Disease risk scoring engine using a comparative machine learning approach evaluating Logistic Regression and Random Forest classifiers, and alert-triggered remote nephrologist consultation, in order to detect silent kidney decline early enough for meaningful clinical intervention, reducing the number of Rwandan adults who reach Stage 4 or Stage 5 Chronic Kidney Disease without warning.

1.3 Project's Main Objective

To develop NephroSasa Rwanda, a pre-dialysis kidney health monitoring web platform that tracks vital readings across multiple clinic visits and home monitoring sessions for Rwandan adults already diagnosed with hypertension or Type 2 diabetes, detects silent kidney decline trajectories before symptoms appear, and connects patients with verified nephrologists for early remote intervention before dialysis becomes necessary.

In this research, "longitudinal" refers specifically to the system's ability to store, retrieve, and analyze a patient's blood pressure and blood sugar readings across time rather than evaluating each reading in isolation. Instead of assessing whether a single visit's reading is dangerous, NephroSasa assesses whether the pattern of readings over multiple visits indicates a worsening kidney health trajectory. This is the core technical distinction between NephroSasa and existing health record systems, which store readings but perform no cross-visit trajectory analysis.

1.3.1 Specific Objectives

- To investigate and analyze the kidney health monitoring gap facing hypertensive and Type 2 diabetic patients in Rwanda by reviewing clinical literature, examining existing system limitations, and establishing the evidence base for longitudinal trajectory-based monitoring as a clinically necessary intervention, completed by May 2026.
- To design and implement a unified longitudinal patient health record system that stores and tracks blood pressure and blood sugar readings across multiple clinic visits and home entries over time, enabling trajectory-based analysis of kidney health for hypertensive and Type 2 diabetic patients in Rwanda, completed by June 2026.
- To design, train, and validate a comparative CKD risk scoring module evaluating Logistic Regression (baseline, 74.70% accuracy) and Random Forest (production, 84.94% accuracy) classifiers using a class-balanced 1,659-record open-source clinical dataset (El Kharoua, Kaggle, 2024), optimized via SMOTE oversampling, and exposed as a Django REST API endpoint returning a low, medium, or high kidney risk score, completed by June 2026.

- To implement an alert-triggered remote nephrologist consultation system integrated with MTN Mobile Money and Airtel Money payment processing and Africa's Talking SMS reminder infrastructure, enabling rural Rwandan patients to receive specialist kidney care remotely when the artificial intelligence model detects a dangerous trajectory, completed by July 2026.

1.4 Research Questions

- Can a comparative machine learning approach evaluating Logistic Regression and Random Forest classifiers achieve a minimum accuracy of 85% with a false-positive rate not exceeding 20%, enabling trajectory-based kidney risk classification across multiple clinic visits?
- Does tracking blood pressure and blood sugar readings longitudinally across multiple visits, rather than evaluating single snapshots, improve the early detection of kidney risk trajectories in the target population?
- Can an alert-triggered remote nephrologist consultation system, integrated with mobile money payment infrastructure, provide clinically meaningful early-stage kidney care access to rural Rwandan patients outside Kigali?

1.5 Project Scope

NephroSasa Rwanda targets Rwandan adults aged 30 and above who are already diagnosed with hypertension or Type 2 diabetes, and does not serve as a detection or diagnostic tool for hypertension, diabetes, or Chronic Kidney Disease itself. Geographic scope for testing is Kigali, Rwanda, using a sample of registered test users.

For the capstone submission, the platform is a web application only. The minimum viable product delivers Layer 1 and Layer 2 in full: unified longitudinal patient health records with manual vital logging, QR code-based health record sharing, the artificial intelligence Chronic Kidney Disease risk scoring engine, automated doctor and patient alerts, and 7-day and 14-day missed follow-up SMS reminders. Layer 3 is partially delivered, covering the remote consultation booking flow and MTN Mobile Money and Airtel Money payment integration. A Flutter offline mobile application for community health workers, Bluetooth device synchronization, and live integration with hospital electronic health record systems are Phase 2 features representing post-graduation development.

1.6 Significance and Justification

NephroSasa Rwanda directly addresses a gap that no existing platform in Rwanda currently fills. By connecting vital readings longitudinally across time, the platform enables early detection of kidney decline in the window where intervention remains clinically possible and financially affordable, before Stage 4 or Stage 5 Chronic Kidney Disease is reached (Nkurunziza et al., 2025).

At RWF 18,000 per year, NephroSasa costs less than a single dialysis session. For Rwandan households for whom RWF 900,000 monthly dialysis costs represent financial catastrophe, NephroSasa makes preventative kidney monitoring economically accessible (Cost Effectiveness and Resource Allocation, 2024). The platform also reduces the geographic barrier to specialist kidney care by connecting rural patients with Kigali-based nephrologists remotely, addressing the severe nephrologist scarcity documented in the International Society of Nephrology Global Kidney Health Atlas (2024), which found Sub-Saharan Africa has the lowest nephrology workforce density globally.

The AstraZeneca Healthy Heart Africa program's expansion into Rwanda in 2025 confirms that international health stakeholders recognize the growing kidney disease burden in Rwanda as a priority gap requiring urgent intervention (CNBC Africa, 2025). NephroSasa aligns with this recognition by providing a locally deployable, mobile-first, AI-powered monitoring solution built specifically for Rwanda's healthcare infrastructure.

1.7 Research Budget

Table 1.1 presents the estimated costs for implementing and deploying NephroSasa Rwanda during the capstone phase.

Table 1.1: NephroSasa Rwanda Capstone Research Budget

Item	Purpose	Cost (RWF)
Supabase Free Plan	Managed PostgreSQL database	0 (Free tier)
Vercel Hobby Plan	Frontend deployment	0 (Free tier)
Render Free Plan	Backend deployment	0 (Free tier)

Africa's Talking API	SMS reminders (pay-per-use)	~5,000
MTN Mobile Money API	Payment processing (sandbox)	0 (Sandbox)
Airtel Money API	Payment processing (sandbox)	0 (Sandbox)
Domain name (optional)	nephrosasa.rw or similar	~30,000
TOTAL		~35,000

1.8 Research Timeline

The Gantt chart below presents the NephroSasa Rwanda development timeline from May to July 2026. Activities actively under development in each month are indicated by the shaded cells.

Table 1.2: NephroSasa Rwanda Development Timeline (May to July 2026)

Activity	May	Jun	Jul
Dataset Cleaning and AI Model Training			
Frontend Rebranding to NephroSasa			
Backend API Development			
New Frontend Pages Development			
AI Endpoint Integration			
Africa's Talking SMS Integration			
Testing: Unit and Integration			
CI/CD Pipeline Setup			
Deployment: Vercel and Render			
Documentation and README			
Final Supervisor Review and Submission			

1.9 Ethical Considerations

NephroSasa Rwanda handles sensitive personal health data, including blood pressure readings, blood sugar levels, kidney risk scores, and patient identity information. This research is conducted in accordance with ethical principles governing health data management and software research.

Patient data privacy is protected through JWT token-based authentication, ensuring that no patient record is accessible without verified credentials. All data is stored on Supabase's managed PostgreSQL infrastructure, which operates under industry-standard encryption at rest and in transit. Patients are the sole owners of their health records, and the QR code sharing feature requires active patient initiation, meaning no health record is shared without explicit patient action.

Informed consent is embedded into the platform's registration flow. Every patient who creates a profile on NephroSasa Rwanda is presented with a plain-language explanation of how their data will be used, specifically that their readings will be analyzed by a logistic regression model to generate a kidney risk score, and that this score is a pattern-detection output, not a clinical diagnosis. The platform does not replace a licensed clinician and makes this clear at every risk score display.

The risk of false positives is acknowledged as an ethical concern. A model that incorrectly flags a patient as high risk may cause unnecessary anxiety or lead to unnecessary medical visits. To mitigate this, NephroSasa targets a false-positive rate not exceeding 20%, all risk scores are presented with a confidence percentage, and every risk result screen includes the disclaimer: "This score is generated by an AI pattern detection model and must be reviewed by a qualified nephrologist before any clinical decision is made."

Doctor verification through the Rwanda Medical and Dental Council number prevents unqualified individuals from accessing patient records under the guise of being a clinician. No doctor account is granted full patient record access until their RMDC number is validated.

This research does not collect data from real patients during the capstone phase. All model training and testing is conducted using the El Kharoua open-source clinical dataset (Kaggle, 2024) and synthetic test user accounts. Should NephroSasa Rwanda proceed to clinical deployment post-graduation, formal ethical clearance from the Rwanda National Ethics Committee and the Rwanda Biomedical Centre will be sought before any real patient data is collected.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This literature review searched for software-related studies addressing kidney health monitoring, machine learning-based Chronic Kidney Disease prediction, longitudinal health record systems, and non-communicable disease management platforms in Sub-Saharan Africa, with a specific focus on Rwanda. Literature was sourced from PubMed, Google Scholar, medRxiv, PLoS Global Public Health, Lancet Global Health, BMC Nephrology, and official reports from the World Health Organization, the International Society of Nephrology, and the Rwanda National Institute of Statistics. Thirteen primary sources were reviewed, all published between 2022 and 2025, consistent with the five-year currency requirement for capstone literature.

2.2 Overview of Existing Systems

Several software systems exist that partially address aspects of the problem NephroSasa targets. However, none addresses the specific gap of longitudinal, pre-symptomatic kidney health monitoring for hypertensive and Type 2 diabetic patients in a low-resource Sub-Saharan African context. Table 2.1 presents the key existing systems reviewed.

Table 2.1: Overview of Existing Systems and Their Limitations

System	Function	Limitation vs NephroSasa
Jordan Nguepi CKD Dashboard (ALU, 2025)	Visualizes CKD progression post-diagnosis	Operates after diagnosis; no pre-diagnosis trajectory monitoring
GLUCOGARD AI App (ALU, 2025)	Diabetes risk prediction and self-management	Predicts diabetes onset; does not monitor kidney impact of existing diabetes
Interactive Health Risk Assessment Tool (ALU, 2025)	ML and chatbot for diabetes and heart disease risk	Targets heart disease; no longitudinal kidney monitoring for hypertensive patients

Epic MyChart / Generic EHR Platforms	Digital health records storage	Store readings but do not analyze longitudinal kidney risk trajectories
KidneyIntelX (US-based)	Biomarker-based CKD progression prediction	Requires laboratory biomarkers; not viable in low-resource Rwandan district clinic settings
Africa-wide mHealth platforms (generic)	General vital logging and reminders	No AI-powered kidney risk scoring; no longitudinal trajectory analysis

2.3 Review of Related Work

2.3.1 Global and Regional Burden of Chronic Kidney Disease

The Lancet Global Health (2024) global kidney disease burden update situates NephroSasa within a worldwide trend: low- and middle-income countries bear a disproportionate share of Chronic Kidney Disease burden precisely because prevention and early monitoring infrastructure is absent. The study estimates that over 70% of global Chronic Kidney Disease cases are attributable to hypertension and diabetes, the exact two conditions NephroSasa targets (Lancet Global Health, 2024).

The International Society of Nephrology Global Kidney Health Atlas (2024) found that Sub-Saharan Africa has the lowest nephrology workforce density globally, with fewer than one nephrologist per one million people in many countries. This finding directly justifies NephroSasa's Layer 3 design decision to enable remote nephrologist consultation for rural patients without requiring physical travel to Kigali (ISN, 2024).

The Rwanda Demographic and Health Survey albuminuria prevalence study provides population-level evidence that kidney damage markers are already present in a significant portion of Rwandan adults with hypertension and diabetes, many of whom are unaware of their early-stage kidney stress. Albuminuria is the earliest detectable sign of kidney damage and is directly linked to uncontrolled hypertension and diabetes.

Together, these three sources establish the global and regional scale of the problem that NephroSasa Rwanda is designed to address at a local level.

2.3.2 The CKD Monitoring Gap in Rwanda

Nkurunziza et al. (2025), in a World Congress of Nephrology study conducted across King Faisal Hospital, the Centre Hospitalier Universitaire de Kigali, and Rwanda Military Hospital between May 2023 and March 2024, documented at least 240 new advanced Chronic Kidney Disease cases annually in Rwanda and explicitly identified the absence of preventive monitoring systems as the central gap in Rwanda's kidney care landscape. This study provides the most direct clinical evidence for the gap NephroSasa addresses.

The BMC Nephrology (2022) study on End-Stage Kidney Disease quality of life at the Centre Hospitalier Universitaire de Kigali and Rwanda Military Hospital documented that patients reaching end-stage kidney disease in Rwanda face catastrophic out-of-pocket costs even under insurance coverage, reinforcing that financial prevention through early monitoring is not merely clinically preferable but economically necessary for Rwandan households (BMC Nephrology, 2022).

The Cost Effectiveness and Resource Allocation (2024) rapid analysis of hemodialysis versus peritoneal dialysis costs in Rwanda provides the specific financial figures cited throughout NephroSasa's business case: RWF 75,000 per dialysis session, three times weekly, totaling approximately RWF 900,000 monthly for uninsured patients (Cost Effectiveness and Resource Allocation, 2024).

The Assessment of Blood Pressure Control Among Hypertensive Patients Attending Rwandan District Hospital Non-Communicable Disease Clinics (2025) is the most operationally relevant study for NephroSasa. It confirms that blood pressure readings are regularly taken at Rwanda's district hospital NCD clinics but are not analyzed longitudinally for kidney risk trajectory, directly validating NephroSasa's core architectural decision (Assessment of Blood Pressure Control Among Hypertensive Patients, 2025).

The PLoS Global Public Health (2025) evaluation of King Faisal Hospital's kidney transplant program documents the post-dialysis patient journey and the high mortality rates associated with late-stage kidney disease detection in Rwanda, further reinforcing the clinical urgency of pre-dialysis intervention that NephroSasa provides (PLoS Global Public Health, 2025).

Collectively, these five sources quantify both the clinical consequences of Rwanda's monitoring gap and the financial catastrophe that early detection could prevent.

2.3.3 Machine Learning Approaches to CKD Detection

Niyonzima et al. (2024), in a medRxiv preprint on machine learning prediction of Chronic Kidney Disease in Rwandan diabetic patients, demonstrated that machine learning models can achieve clinically meaningful accuracy in predicting kidney disease progression in a Rwandan patient population, providing Rwanda-specific validation context for the comparative machine learning approach used in NephroSasa. This is significant given that the University of California Irvine Chronic Kidney Disease Dataset used to train NephroSasa's model was sourced from hospital records in India (UCI Machine Learning Repository, 2015), and Rwanda-specific validation literature is therefore essential for establishing the model's clinical applicability.

This Rwanda-specific validation study directly supports the comparative machine learning methodology adopted in NephroSasa, bridging the gap between the open-source training dataset and the Rwandan clinical context.

2.3.4 Existing Platforms and Their Limitations

The AstraZeneca Healthy Heart Africa Chronic Kidney Disease expansion into Rwanda (CNBC Africa, 2025) signals that international health stakeholders are actively investing in Rwanda's kidney disease gap, validating NephroSasa's market timing and relevance.

This investment confirms that international stakeholders recognize Rwanda's CKD gap as urgent, and that no existing platform currently addresses pre-diagnosis longitudinal monitoring for the target population.

2.3.5 Summary of Reviewed Literature

Table 2.2 presents a structured summary of all thirteen reviewed sources.

Table 2.2: Summary of Reviewed Literature

Author(s) / Source	Year	Relevance to NephroSasa
Rwanda National Institute of Statistics	2024	Confirms NCDs account for 47.7% of hospital deaths in Rwanda
Nkurunziza et al.	2025	Found 240+ new advanced CKD cases annually; calls for preventive measures
Niyonzima et al. (medRxiv)	2024	Rwanda-specific ML validation for CKD prediction models
PLoS Global Public Health	2025	Documents post-dialysis outcomes and cost burden in Rwanda
Lancet Global Health	2024	Over 70% of global CKD cases attributable to hypertension and diabetes
BMC Nephrology	2022	Catastrophic financial cost of end-stage kidney disease in Rwanda
ISN Global Kidney Health Atlas	2024	Sub-Saharan Africa has lowest nephrology workforce density globally
RDHS	Recent	Albuminuria prevalence in Rwandan adults with hypertension and diabetes
WHO Rwanda NCD Profile	Current	Official 16.8% hypertension and 2.9% diabetes prevalence figures
Cost Effectiveness and Resource Allocation	2024	RWF 75,000 per dialysis session quantified for Rwanda

CNBC Africa / AstraZeneca	2025	Industry investment confirming Rwanda CKD gap recognized globally
Assessment of BP Control, District NCD Clinics	2025	Confirms readings taken but not longitudinally analyzed for kidney risk
UCI Machine Learning Repository	2015	Originally specified training dataset; replaced by El Kharoua (2024) 1,659-record open-source clinical dataset during model development for improved robustness

2.4 Strengths and Weaknesses of Existing Systems

The primary strength of existing systems such as Jordan Nguenzi's 2025 ALU capstone CKD visualization dashboard is the quality of post-diagnosis patient engagement. However, the fundamental weakness is that these systems begin too late: they require a formal diagnosis before they can function, meaning they offer no value during the Stage 1 to Stage 3a prevention window where NephroSasa operates.

Generic mHealth platforms provide patients with a record of their readings but do not apply artificial intelligence to detect kidney risk trajectories. They are data storage tools, not risk intelligence tools. The absence of longitudinal analysis means they cannot detect the slow, multi-month trends that signal approaching kidney failure. The weakness shared by all reviewed systems is the absence of longitudinal artificial intelligence kidney risk scoring for the pre-diagnosis population of hypertensive and Type 2 diabetic patients.

2.5 General Comment and Conclusion

The literature consistently confirms three findings that directly justify NephroSasa Rwanda. First, hypertension and Type 2 diabetes are the dominant drivers of Chronic Kidney Disease in Rwanda and globally (Lancet Global Health, 2024). Second, detection in Rwanda consistently occurs at Stage 4 or Stage 5, where dialysis is the only remaining option and costs are catastrophic (Nkurunziza et al., 2025). Third, no existing software platform, whether at African Leadership University or in the broader Rwandan health

technology landscape, monitors kidney health longitudinally for hypertensive and Type 2 diabetic patients before symptoms appear. NephroSasa Rwanda is positioned at the intersection of these three findings.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

NephroSasa Rwanda is designed using an agile incremental development model. The system is built in three layers, each independently functional but connected: Layer 1 for longitudinal health records, Layer 2 for artificial intelligence risk scoring, and Layer 3 for alert-triggered remote care. The backend uses Django and Django REST Framework. The frontend uses Next.js 14 with TypeScript and TailwindCSS. The database is PostgreSQL managed through Supabase. The artificial intelligence module is built in Python using scikit-learn, evaluating a Logistic Regression baseline and a Random Forest production model trained on a 1,659-record open-source clinical dataset (El Kharoua, Kaggle, 2024).

3.2 Research Design and Development Model

NephroSasa Rwanda follows an agile incremental development approach. The system is built in clearly defined increments: the AI model is trained and validated first, followed by the backend API endpoints, followed by the frontend pages, followed by integration and deployment. Each feature is built end-to-end (database, backend, frontend, test) before the next feature begins. This approach is chosen because NephroSasa's core value, longitudinal AI risk scoring, cannot be demonstrated without all three layers functioning together.

3.2.1 Data Sampling Strategy

The primary dataset used to train NephroSasa Rwanda's kidney risk scoring model is an open-source chronic kidney disease clinical repository comprising 1,659 patient records across 10 clinical features, compiled by Rabie El Kharoua and published on Kaggle (2024). This dataset was adopted in place of the originally specified UCI Chronic Kidney Disease Dataset (400 records) following a supervised review that identified the UCI dataset as insufficiently sized for robust model training. The 1,659-record dataset contains all core clinical indicators relevant to hypertension- and diabetes-induced kidney decline, including systolic and diastolic blood pressure, fasting blood sugar, HbA1c, serum creatinine, GFR, and BUN levels.

As disclosed in Section 1.9, real patient data is not collected or processed during this capstone phase. The training dataset is an open-source repository used strictly for software prototyping and model validation, consistent with standard medical informatics engineering practice.

To address severe class imbalance in the original dataset (135 healthy patients versus 1,524 CKD patients), a SMOTE pipeline was applied to the training split, producing a balanced training set of 2,438 records (1,219 per class). Two models were evaluated: a Logistic Regression baseline achieving 74.70% accuracy with a 22.22% false positive rate, and a Random Forest production model achieving 84.94% accuracy. The Random Forest model was selected as the production engine and serialized as `nephrosasa_model.pkl` for integration with the Django REST API endpoint. Cross-validation across five stratified folds yielded a mean accuracy of 79.12% with a standard deviation of 1.36%, confirming model stability. Future post-graduation deployment will involve retraining on Rwanda-specific patient data following RNEC and RBC ethical clearance.

3.3 Class Diagram

The primary classes in NephroSasa Rwanda are Patient, VitalLog, KidneyRiskScore, Doctor, Consultation, and SMSReminder. The Patient class holds demographic data and diagnosis type. The VitalLog class records blood pressure and blood sugar readings with timestamp and source. The KidneyRiskScore class stores the AI model output (low, medium, or high) with confidence value and timestamp. The Doctor class holds Rwanda Medical and Dental Council verification status and specialty. The Consultation class links a Patient to a Doctor for a remote session with booking time and payment status. The SMSReminder class tracks reminder dispatch status and trigger type (7-day or 14-day).

Figure 3.1: Class Diagram

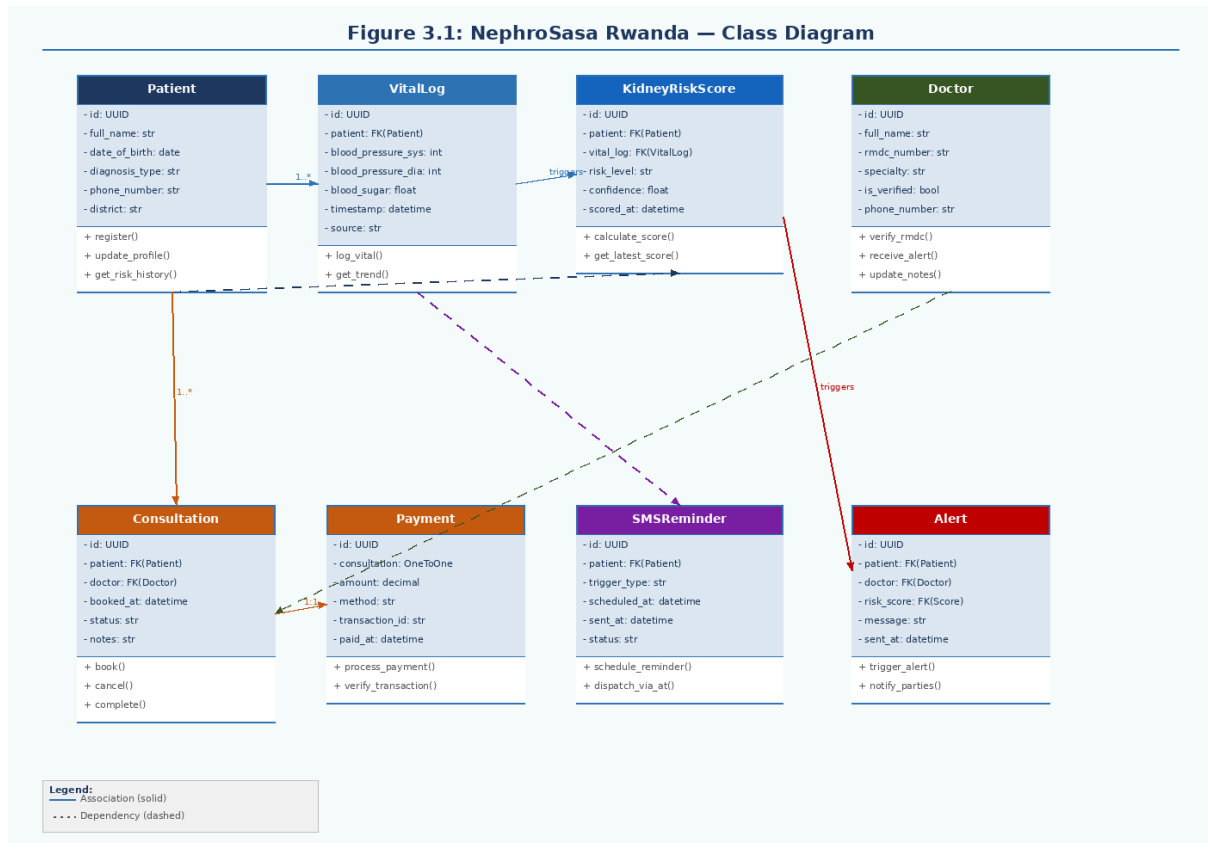


Figure 3.1 — NephroSasa Rwanda Class Diagram showing primary classes, attributes, methods, and relationships

3.4 System Architecture Diagram

The architecture follows a three-tier pattern: presentation tier (Next.js frontend on Vercel), application tier (Django REST Framework backend on Render with the scikit-learn AI module), and data tier (PostgreSQL via Supabase). Africa's Talking API handles outbound SMS reminders. MTN Mobile Money and Airtel Money APIs handle payment processing. All communication between frontend and backend uses HTTPS REST API calls with JWT authentication.

Figure 3.2: System Architecture Diagram

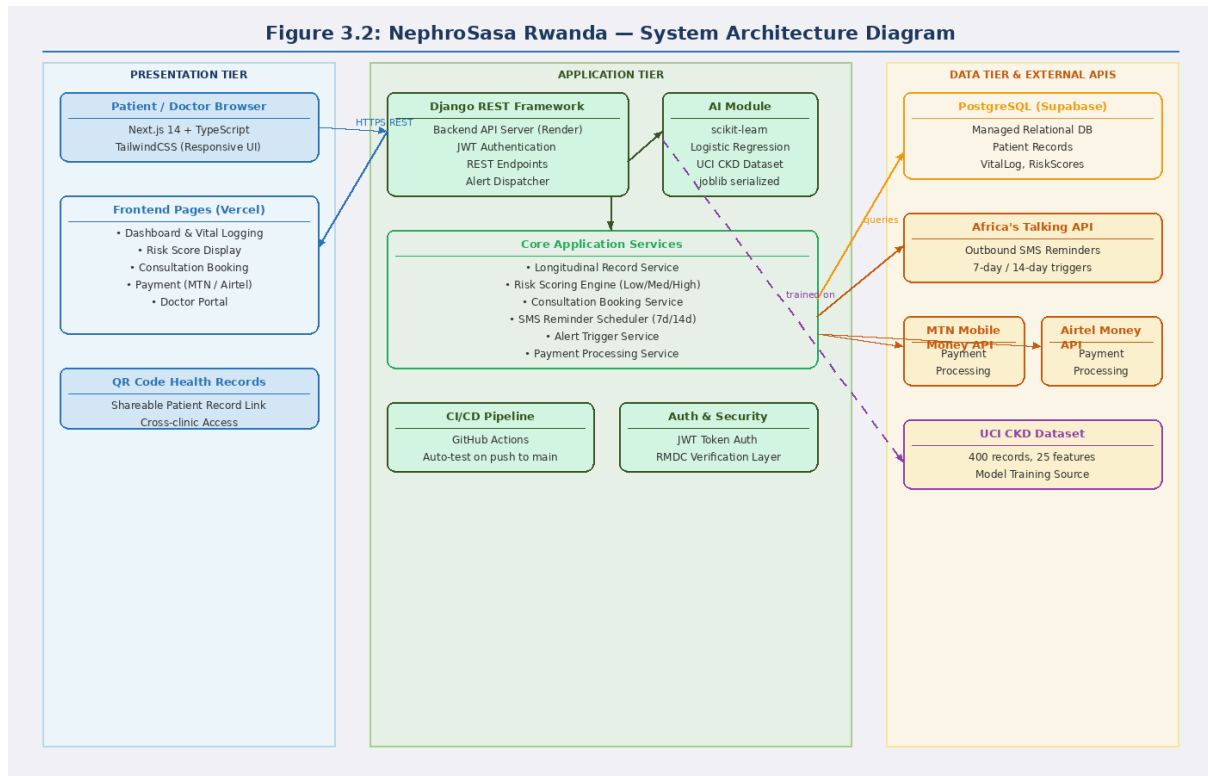


Figure 3.2 — NephroSasa Rwanda Three-Tier System Architecture (Presentation, Application, Data)

3.5 UML Diagrams: Entity Relationship Diagram

The NephroSasa Rwanda database has the following primary entities and relationships. A Patient has many VitalLog entries (one-to-many). A Patient has many KidneyRiskScore records over time (one-to-many). A Patient can book many Consultations (one-to-many). A Doctor can receive many Consultations (one-to-many). A Consultation has one Payment record (one-to-one). A Patient has many SMSReminder records (one-to-many). The VitalLog table is the central data source for the AI risk scoring engine: each new VitalLog entry triggers a recalculation of the patient's KidneyRiskScore.

Figure 3.3: Entity Relationship Diagram

Figure 3.3: NephroSasa Rwanda — Entity Relationship Diagram

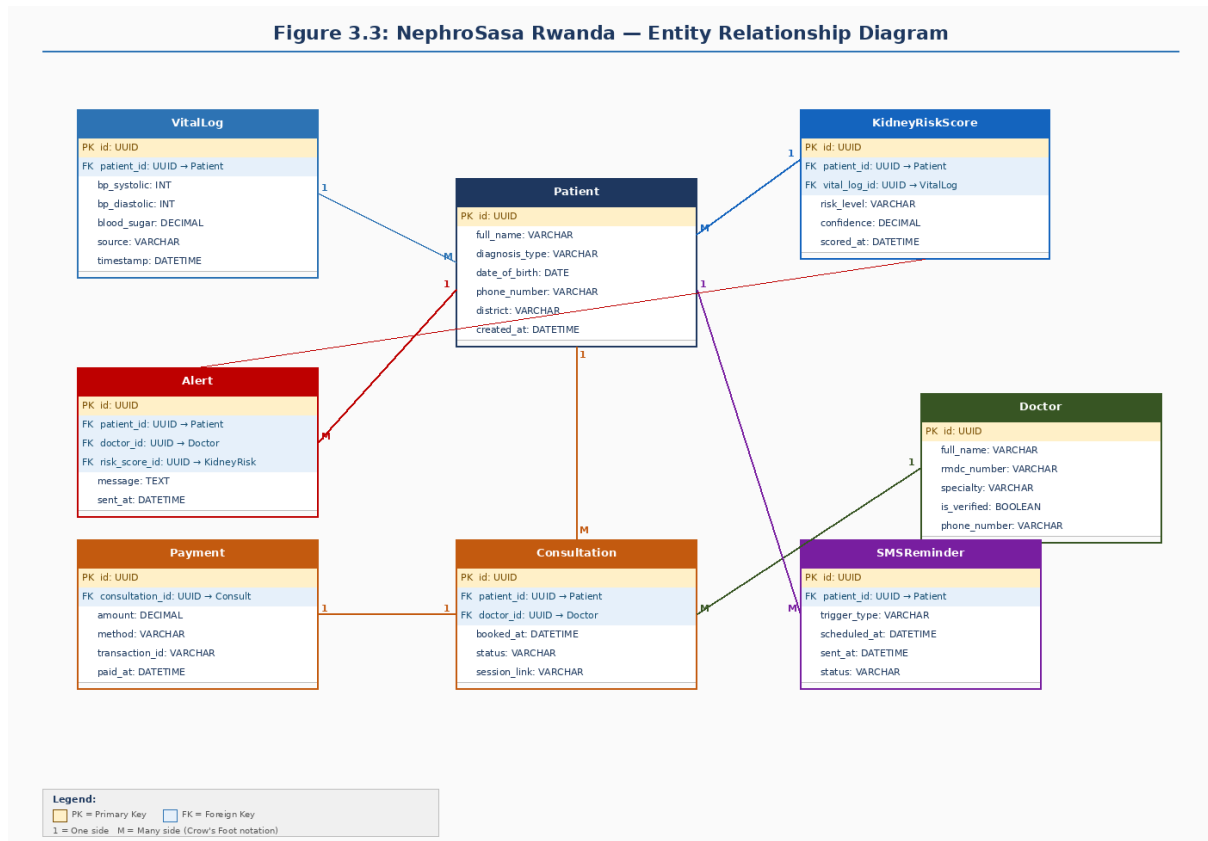


Figure 3.3 — NephroSasa Rwanda Database Entity Relationship Diagram with cardinalities

3.6 Use Case Diagram

NephroSasa Rwanda has three primary actors: the Patient, the Doctor (Nephrologist), and the System (automated processes). The Patient can register, log vitals, view their kidney risk score, receive SMS reminders, book a remote consultation, and make payment via mobile money. The Doctor can register with Rwanda Medical and Dental Council verification, view a patient's full longitudinal health record, update consultation notes, and receive automated alerts when a patient's kidney risk score deteriorates. The System automatically scores kidney risk after each new vital entry, sends 7-day and 14-day SMS reminders to patients who have not logged vitals, and sends alerts to both patient and doctor when a dangerous trajectory is detected.

Figure 3.4: Use Case Diagram

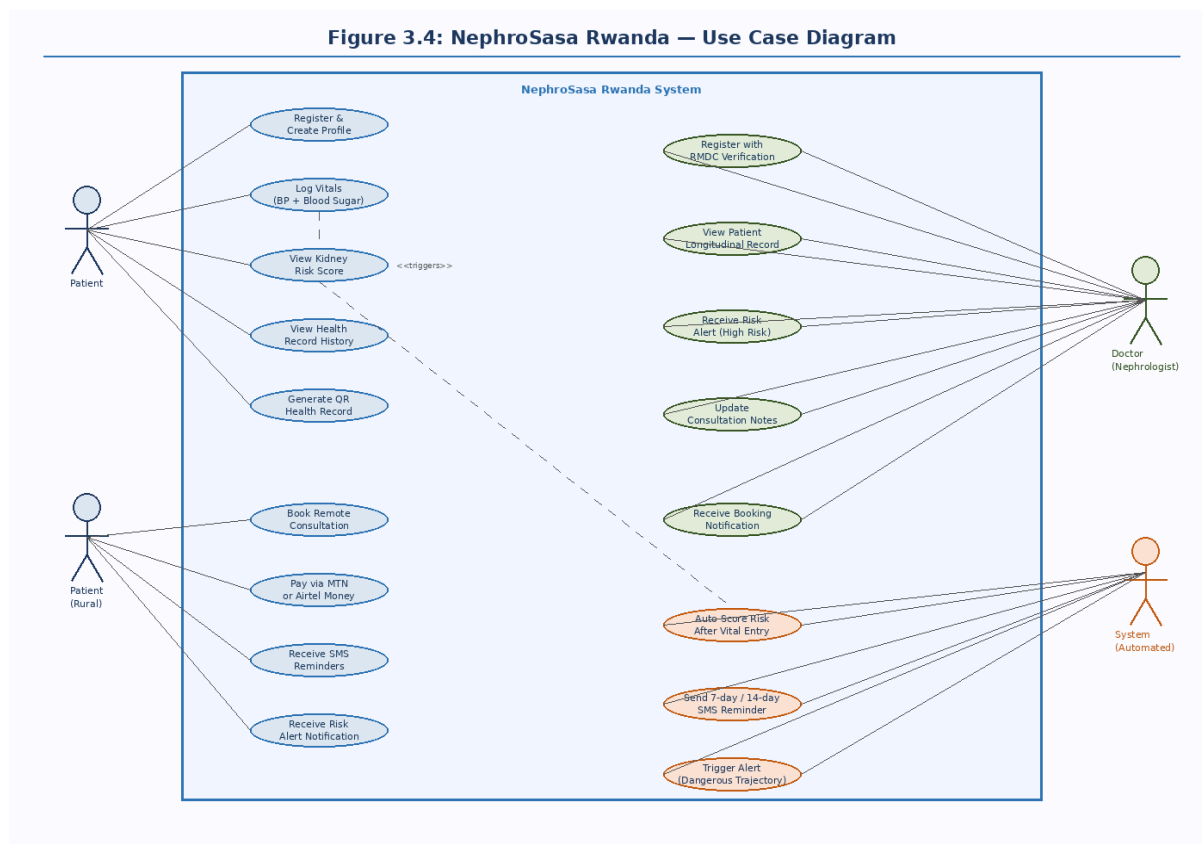


Figure 3.4 — NephroSasa Rwanda Use Case Diagram (Patient, Doctor, System actors)

3.7 Development Tools

Table 3.1 presents the complete technology stack for NephroSasa Rwanda.

Table 3.1: NephroSasa Rwanda Technology Stack

Layer	Technology	Purpose
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Frontend	Next.js 14 with TypeScript	Web application framework with server-side rendering
Frontend	TailwindCSS	Mobile-first responsive styling
Frontend	Vercel	Frontend deployment and CDN
Backend	Django and Django REST Framework	API server and longitudinal database query support
Backend	Render	Backend deployment
Database	PostgreSQL via Supabase	Managed relational database for longitudinal records
AI Module	Python and scikit-learn	Random Forest production model and Logistic Regression baseline training and inference
AI Module	joblib	Model serialization and loading
AI Module	pytest	Unit testing for AI model
Alerts	Africa's Talking API	SMS reminders at 7-day and 14-day intervals
Payments	MTN Mobile Money API	Mobile payment processing
Payments	Airtel Money API	Mobile payment processing
CI/CD	GitHub Actions	Automated testing on every push to main branch
Version Control	Git and GitHub	Source control and CI/CD trigger

References

- AstraZeneca. (2025, September). Healthy Heart Africa Chronic Kidney Disease expansion in Rwanda. CNBC Africa.
<https://www.cnbc africa.com/media/7758018035387/astrazeneca-expands-ncd-care-to-include-chronic-kidney-disease-in-rwanda-/>
- Ntaganda, E., et al. (2025). Assessment of blood pressure control among hypertensive patients attending Rwandan district hospital Non-Communicable Disease clinics. Rwanda Journal of Medicine and Health Sciences, 8(1).
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11790538/>
- BMC Nephrology. (2022). End-Stage Kidney Disease quality of life study across the Centre Hospitalier Universitaire de Kigali and Rwanda Military Hospital. BMC Nephrology. <https://bmcnephrol.biomedcentral.com/articles/10.1186/s12882-022-02958-6>
- Cost Effectiveness and Resource Allocation. (2024). Rapid cost-effectiveness analysis of hemodialysis versus peritoneal dialysis in Rwanda.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11059575/>
- International Society of Nephrology. (2024). Global Kidney Health Atlas 2024.
<https://www.theisn.org/global-kidney-health-atlas>
- Lancet Global Health. (2024). Global kidney disease burden and care update. The Lancet Global Health, 12(1).
[https://www.thelancet.com/journals/langlo/article/PIIS2214-109X\(24\)00050-0/fulltext](https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(24)00050-0/fulltext)
- Niyonzima, N., et al. (2024). Machine learning prediction of Chronic Kidney Disease in Rwandan diabetic patients. medRxiv.
<https://www.medrxiv.org/content/10.1101/2024.10.23.24315998v1>

Nkurunziza, A., et al. (2025). Chronic Kidney Disease perspectives in Rwanda: A study across King Faisal Hospital, Centre Hospitalier Universitaire de Kigali, and Rwanda Military Hospital (May 2023 to March 2024). World Congress of Nephrology 2025 Proceedings.

PLoS Global Public Health. (2025). King Faisal Hospital kidney transplant program evaluation. PLOS Global Public Health, 5(1).

<https://journals.plos.org/globalpublichealth/article?id=10.1371/journal.pgph.0004905>

Rwanda National Institute of Statistics. (2024). Rwanda Vital Statistics Report 2024. Government of Rwanda. <https://www.statistics.gov.rw>

Rwanda Biomedical Centre. (2020). Rwanda Demographic and Health Survey 2019-20: Key Indicators. Government of Rwanda. <https://www.statistics.gov.rw>

UCI Machine Learning Repository. (2015). Chronic Kidney Disease Dataset (400 records, 25 features). University of California Irvine.

https://archive.ics.uci.edu/ml/datasets/chronic_kidney_disease

El Kharoua, R. (2024). Chronic Kidney Disease Dataset (1,659 records). Kaggle.

<https://www.kaggle.com/datasets/rabieelkharoua/chronic-kidney-disease-dataset-analysis>

World Health Organization. (2023). Rwanda Non-Communicable Disease Country Profile 2023.

<https://www.who.int/teams/noncommunicable-diseases/surveillance/data/rwanda>